NATIONAL UNIVERSITY OF SCIENCES AND TECHNOLOGY (NUST)

School of Mechanical and Manufacturing Engineering (SMME)

Department of Mechanical Engineering

Design, Analysis, and Prototyping of a Static Sector-Gate Storm Surge Barrier

Design Firm:

HATech Engineering

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Course Title:	Mechanics of Materials II (MOM-II)
Instructor:	Prof. Aamir Mubashar
Date of Submission:	15 April 2026

Statement on Use of Artificial Intelligence

HATech Engineering acknowledges the use of Claude Sonnet 4.6 (Anthropic) as the designated AI team member for this project, as permitted and encouraged by the project brief. The following statement clarifies the nature and boundaries of this use, in full accordance with Prof. Aamir Mubashar's guidelines.

AI Was Used For	AI Was NOT Used For
✓ Grammar and language polishing	✗ Generating design ideas or concepts
✓ Literature search assistance	✗ Performing or replacing MathCAD calculations
✓ Clarifying structural concepts	✗ Making engineering decisions
✓ Reviewing report tone and clarity	✗ Creating CAD drawings or renderings
✓ Structuring the report layout	✗ Replacing any human analysis or judgment

The intellectual content of this report, the conceptual designs, engineering decisions, analytical approach, calculations, and conclusions, is the original work of the four human team members. Claude was the icing on the top.

Executive Summary

This report presents the complete design, structural analysis, and prototyping documentation for a static sector-gate storm surge barrier, developed by HATech Engineering as part of the Mechanics of Materials II (MOM-II) course at the National University of Sciences and Technology (NUST), Spring 2026.

Inspired by the Maeslantkering storm surge barrier in the Netherlands, the project required the design and fabrication of a 600 mm cantilever structure capable of sustaining a horizontal resultant force of 50 N for at least 10 seconds, using only balsa wood or ice cream sticks, or cardboard, and glue. The structure converges to a single 10 mm diameter pivot anchor.

Three conceptual designs were systematically evaluated: (A) a single central spine with lateral cross-bracing, (B) a dual lattice arm configuration, and (C) a triangulated truss frame with a curved gate panel. A weighted decision matrix incorporating seven criteria, structural efficiency, load path clarity, ease of fabrication, material utilisation, serviceability, aesthetics, and prototype risk, was employed to select the optimal design. Design C (Triangulated Truss Frame) emerged as the preferred solution.

Analytical evaluation was conducted using MathCAD, covering bending stress, shear stress, buckling analysis, combined loading conditions, and deflection (sag) calculations. Balsa wood was selected as the primary structural material owing to its exceptional specific strength, used at the pivot and all An analytical evaluation was conducted using MathCAD, covering bending and shear stresses high-stress nodes.

The key metric for success is the ratio of Actual Failure Load to Prototype Weight. HATech Engineering's design strategy prioritizes structural efficiency and load-path optimization to maximize this ratio. Detailed CAD drawings, renderings, analytical calculations, and team minutes are included in subsequent sections.

1. Introduction

1.1 Background and Motivation

Coastal flooding and storm surges are among the most destructive natural hazards facing low-lying regions worldwide. Engineered storm surge barriers, hydraulic structures designed to block tidal and storm-driven water intrusion, have proven critical in protecting densely populated coastal zones. The Maeslantkering barrier at Hook of Holland, Netherlands, stands as a defining exemplar: its two enormous, curved sector gates, each spanning 210 meters, are supported by steel lattice arms that converge at a single ball-and-socket joint, transferring the full hydrostatic load to a fixed foundation.

This project abstracts and scales that engineering challenge into a laboratory exercise. Teams are required to design, analyze, and fabricate a static (non-moving) sector-gate barrier at 1:N scale using low-cost, readily available materials, balsa wood, ice cream sticks, cardboard, and glue, under strict geometric and loading constraints. The exercise demands the integration of stress analysis, buckling theory, material selection, failure prediction, and iterative prototyping, reflecting the full cycle of mechanical engineering design practice.

1.2 Objectives and Scope

The objectives of this project are as follows:

- To design a static, self-supporting cantilever sector gate that transfers a horizontal resultant force of 50 N from a curved gate surface to a single fixed pivot point.
- To perform rigorous analytical evaluation of structural members under bending, shear, combined loading, and buckling using MathCAD.
- To evaluate and select from multiple conceptual designs using a systematic decision matrix.
- To construct a physical prototype consistent with the finalized design and within the prescribed geometric envelope.
- To document the complete design process, analysis, and results in a professional technical report.

The scope is limited to static structural analysis. Dynamic loading, fluid-structure interaction, and the gate's operational mechanism are outside the scope of this project.

1.3 HATech Engineering - Design Firm Overview

HATech Engineering is a student design firm established for this project, comprising three members with clearly defined roles:

Role	Member	Primary Responsibility
Project Manager	Muhammad Hamza	Project coordination, scheduling, report compilation, and conceptual design development
Head of Analysis & Material Selection	Muhammad Akshaf	Conceptual design development, Structural analysis, MathCAD calculations, material property research
Head of Design	Muhammad Taha Farrukh Saeed	Conceptual design development, CAD drawings, 3D rendering, prototype fabrication
Head of Systems Engineering	Haissam Bin Zia	Conceptual design development, prototype fabrication

2. Literature Review

2.1 Storm Surge Barriers - Engineering Principles

Storm surge barriers are hydraulic structures designed to prevent tidal or storm-driven floodwater from inundating protected areas. They must withstand significant hydrostatic pressure while maintaining structural integrity under sustained loading. The fundamental engineering challenge lies in the efficient transfer of distributed pressure loads across a curved gate surface to concentrated anchor points, minimizing material use while maximizing load-carrying capacity.

The hydrostatic pressure distribution on a vertical or inclined gate face is triangular in nature, with pressure $\mathbf{p(z) = \rho g h(z)}$, where ρ is the fluid density, g the gravitational acceleration, and h(z) the water depth at height z. The resultant force FR acts at one-third of the water depth from the base, and its magnitude is given by $\mathbf{FR = \tfrac{1}{2}\rho g H^2 B}$, where H is the water height and B is the gate width.

2.2 Structural Analysis Methods

For cantilever structures of this type, classical beam theory (Euler-Bernoulli) provides the analytical basis for bending stress and deflection calculations. For slender compression members, Euler's buckling formula governs the critical load, while Johnson's formula applies in the intermediate slenderness range. Combined loading analysis, accounting for simultaneous bending and compression, employs interaction equations from standard references such as Hibbeler's Mechanics of Materials and Roark's Formulas for Stress and Strain.

The critical buckling load for a pin-ended column is $Pcr = \pi^2 EI / (KL)^2$, where E is the modulus of elasticity, I is the second moment of area, K is the effective length factor, and L is the member length. For balsa wood members of rectangular cross-section, $I = bh^3/12$. Safety factors for wood structures are typically higher than for metals, reflecting greater variability in material properties.

2.3 Balsa Wood as a Structural Material

Balsa wood is widely used in scale models and lightweight structural applications owing to its exceptionally high specific strength (strength-to-density ratio). Its cellular structure provides reasonable compressive and tensile capacity along the grain direction, with significantly lower properties across the grain. At densities of 120–160 kg/m^3, balsa achieves tensile strengths of approximately 20–30 MPa along the grain, making it one of the most structurally efficient natural materials available.

Key considerations in balsa wood design include: (i) significant anisotropy, properties along and across the grain differ by factors of 5–10; (ii) sensitivity to moisture content; (iii) low modulus compared to engineered materials, making deflection rather than stress the often governing design criterion; and (iv) susceptibility to stress concentrations at joints and notches.

2.4 Analogous Scale Model Studies

Several published studies on balsa wood structural model competitions (e.g., ASCE structural competition, AISC student design competitions) confirm that optimized triangulated truss configurations consistently outperform simple beam or spine architectures in load-to-weight efficiency. Key design principles emerging from these studies include maximizing member slenderness within buckling limits, concentrating material at high-stress zones, and using secondary members to prevent out-of-plane instability.

2.5 Summary

The literature review establishes that (i) triangulated truss configurations offer the best structural efficiency for cantilever applications; (ii) balsa wood is an appropriate material for this scale model, provided anisotropy is carefully managed; and (iii) classical analytical methods, beam theory and Euler buckling, are sufficient and appropriate for this analysis, consistent with the requirement that FEA not be used.

3. Conceptual Design

3.1 Design Requirements Summary

Before generating concepts, the design team formally documented the functional and geometric requirements as a basis for comparative evaluation:

- Total reach: 600 mm ± 5 mm (pivot center to gate face)
- Gate height: 200 mm ± 5 mm
- Gate radius: 500 mm ± 5 mm
- Anchor: single 10 mm diameter dowel pivot
- Design load: 50 N horizontal resultant force, sustained 10 seconds
- Maximum sag: 6 mm under dead weight
- Materials: balsa wood, glue only
- No secondary support or feet permitted under arms or gates

3.2 Design A — Single Central Spine

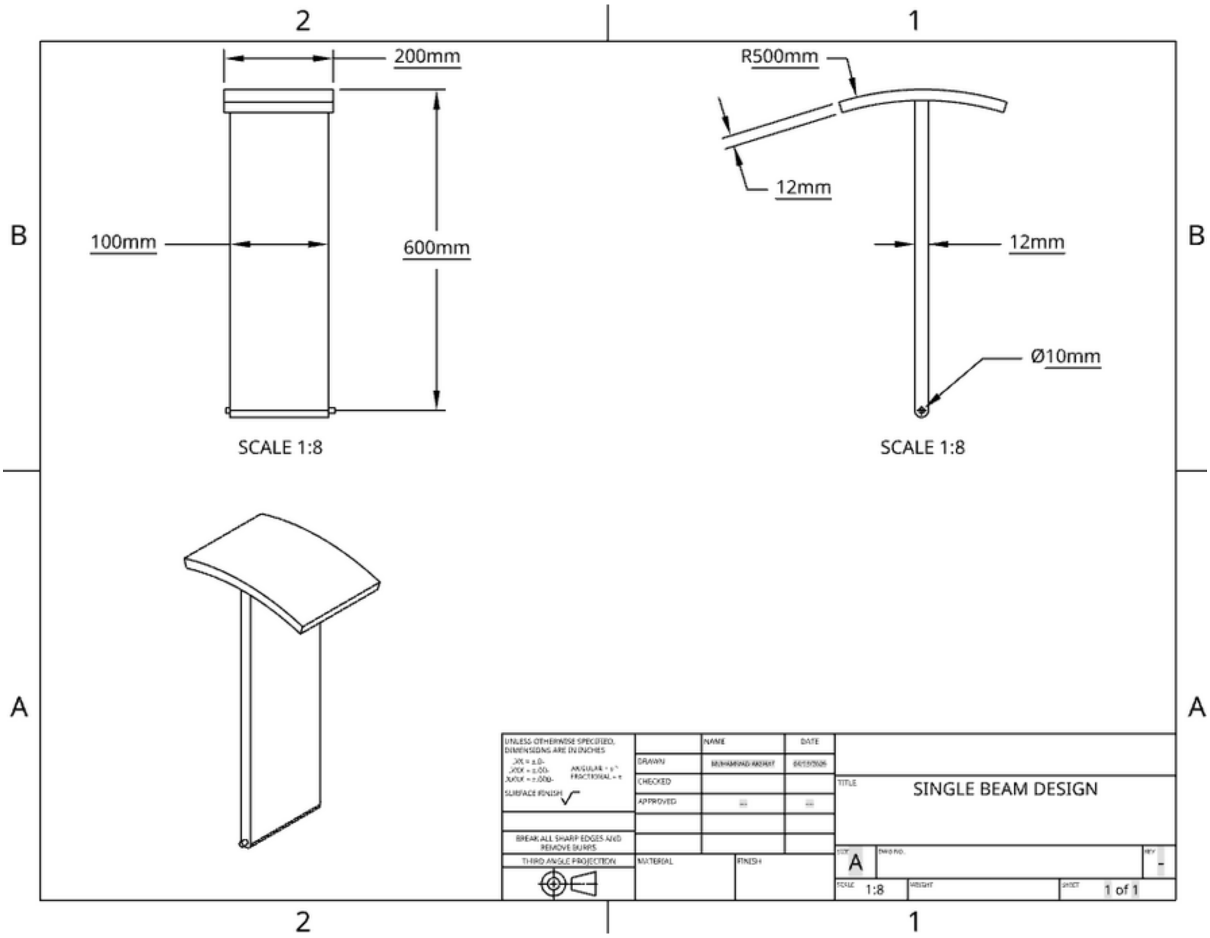


Figure 3.1 — Design A: Single Central Spine with Lateral Cross-Bracing (Concept Sketch)

Design A employs a single central longitudinal member (the 'spine') oriented from the pivot to the gate face, with transverse cross-braces and lateral struts radiating outward to support the curved gate panel. The gate is formed from a bent cardboard or thin balsa sheet, curved to the specified 500 mm radius.

Structural Principle: The spine acts as a cantilever beam in bending, with the gate load applied at its tip. Cross-braces prevent lateral spine buckling. The load path is direct: gate panel → spine tip → spine bending → pivot reaction.

Advantages: Simple geometry, straightforward to fabricate, clear load path, minimal number of members.

Disadvantages: The spine is subjected to large bending moments over its full length, requiring significant cross-sectional area (and thus weight) to prevent excessive deflection and stress. Lateral stability relies entirely on cross-bracing, which adds parasitic weight. The sag criterion (≤ 6 mm) is difficult to satisfy without a significantly oversized spine cross-section.

3.3 Design B — Dual Lattice Arm Configuration

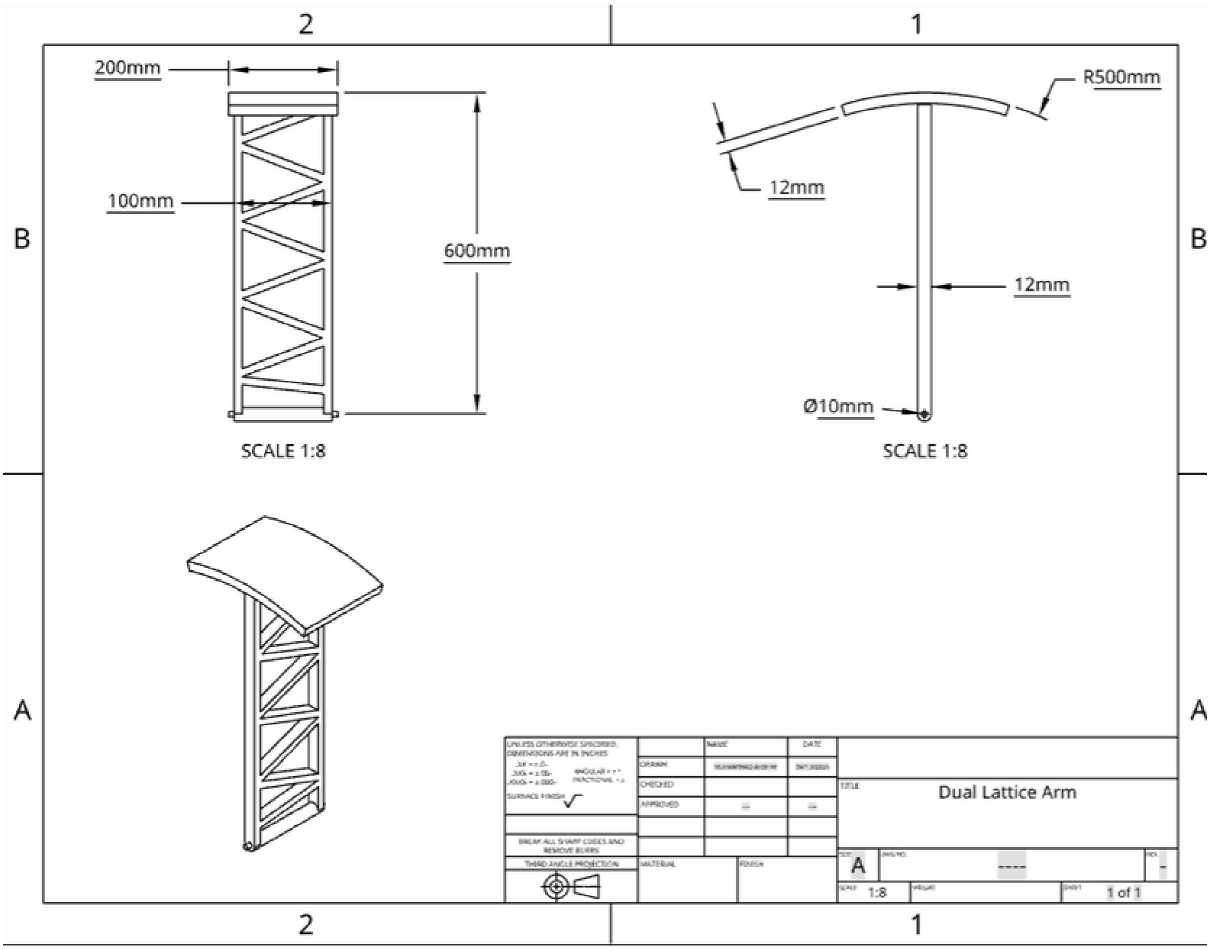


Figure 3.2 — Design B: Dual Lattice Arm Configuration (Concept Sketch)

Design B mimics the Maeslantkering's structural principle at scale. Two lattice arms, each composed of a top chord, a bottom chord, and diagonal web members, converge at the pivot. The curved gate panel is attached at the outer ends of both arms, spanning vertically between them.

Structural Principle: The lattice arms function as deep trusses in bending. The top chord is in tension, the bottom chord in compression, and the diagonals carry shear as axial forces. This is structurally far more efficient than a solid or hollow beam of the same depth, as material is concentrated at the extreme fibres (chords).

Advantages: High structural efficiency, good vertical stability from the two-arm geometry, effective at controlling sag due to distributed member stiffness.

Disadvantages: Greater fabrication complexity, many joint connections (potential for glue joint failure), requires careful alignment during assembly. Out-of-plane stability between the two arms must be ensured.

3.4 Design C — Triangulated Truss Frame with Curved Gate Panel

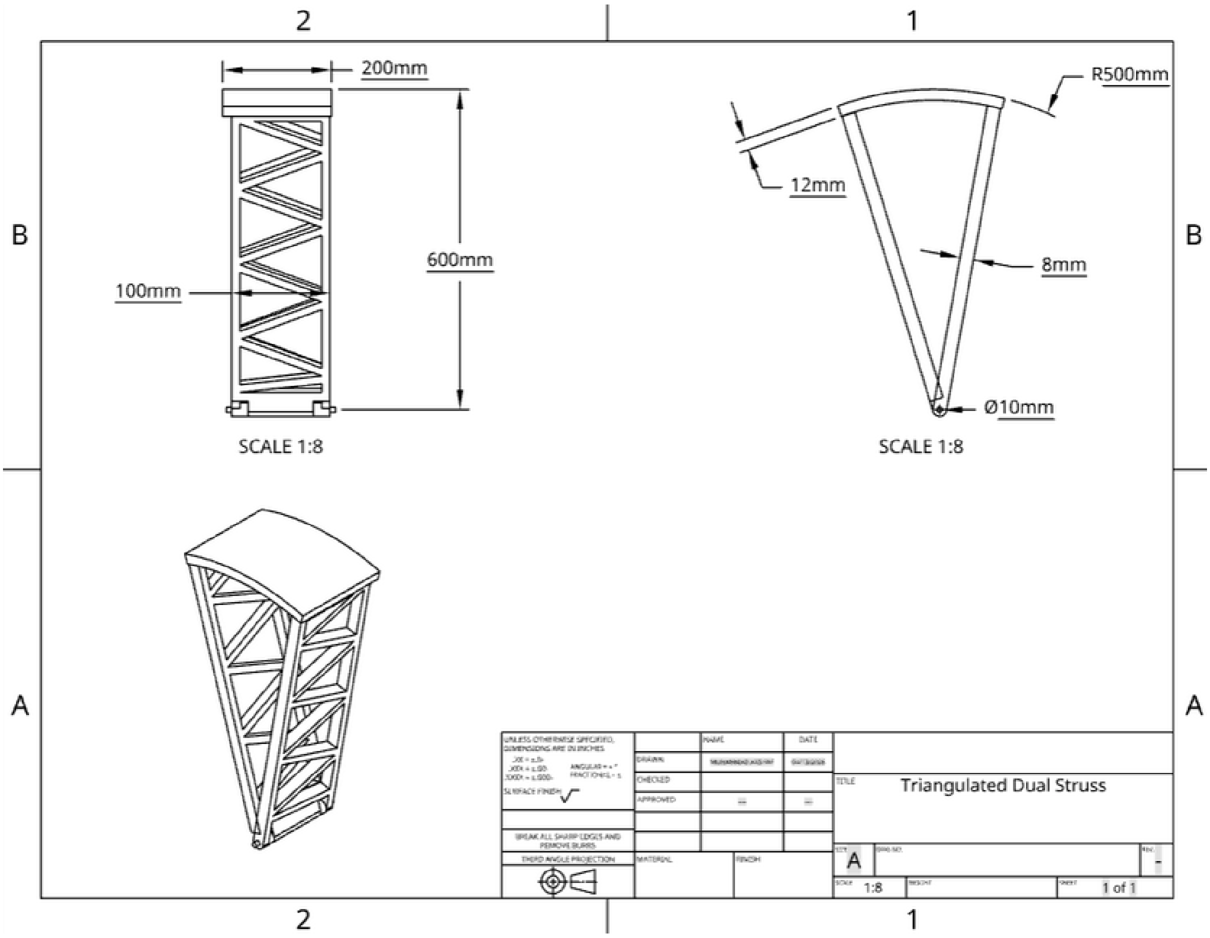


Figure 3.3 — Design C: Triangulated Truss Frame with Curved Gate Panel (Concept Sketch)

Design C employs a fully triangulated planar truss frame in both the horizontal (plan) and vertical (elevation) planes. The horizontal plane truss transfers the gate load to the pivot through a series of diagonal members, while vertical truss elements resist the self-weight bending and sag. The gate panel (cardboard, curved to R = 500 mm) is attached to the outer vertical face.

Structural Principle: Triangulation converts all member forces into pure axial forces (tension or compression), eliminating bending in individual members. This maximizes structural efficiency for a given material volume. The 3D truss geometry provides inherent out-of-plane stability.

Advantages: Maximum structural efficiency (all members in pure axial loading), inherent 3D stability, optimal sag performance, high load-to-weight potential. Well-established in literature as the superior configuration for balsa wood competitions.

Disadvantages: The most complex fabrication of the three designs requires precise cutting and alignment of many members, and more joints (glue bonds) to manage.

3.5 Design Comparison Summary

Table 3.1 provides a qualitative side-by-side comparison of the three designs before formal weighted scoring:

Attribute	Design A Spine + Cross	Design B Lattice Arms	Design C Truss Frame
Structural Efficiency	Moderate	High	Very High
Fabrication Complexity	Low	Moderate	High
Sag Performance	Weak	Good	Excellent
Load Path Clarity	Clear (1D)	Clear (2D)	Complex (3D)
Joint Count	Low (~10)	Medium (~20)	High (~30+)
Out-of-Plane Stability	Poor	Good	Excellent

Table 3.1 — Qualitative Comparison of the Three Conceptual Designs

4. Identification of Critical Parameters

4.1 Critical Design Parameters

The team identified seven critical parameters for evaluating all three designs. The selection process considered the project requirements, testing protocol (UTM loading), and the primary success metric (failure load/prototype weight).

No.	Parameter	Weight	Justification
1	Structural Efficiency	25%	Directly governs the failure-load-to-weight ratio.
2	Serviceability (Sag ≤ 6 mm)	25%	A pass/fail test criterion
3	Load Path Clarity	15%	Clear, direct load paths minimize secondary stresses and joint demands.
4	Ease of Fabrication	15%	Poor fabrication increases geometric imperfections, glue joint eccentricity, and failure risk.
5	Material Utilisation	15%	Efficient material use reduces dead weight, improving the success metric.
6	Buildability / Prototype Risk	5%	Designs that are difficult to build precisely introduce geometric imperfections, degrading performance.
—	TOTAL	100%	—

Table 4.1 — Critical Design Parameters and Weightings

4.2 Weighting Rationale

Structural efficiency and serviceability together account for 40% of the total weight, reflecting the fact that these two criteria directly map to the pass/fail testing requirements and the primary scoring metric. Load path clarity and ease of fabrication are weighed equally at 15% each, recognising that an analytical sound design that cannot be built accurately will underperform. Material utilisation is also 15%, reflecting the denominator in the success metric formula. Innovation and buildability risk round out the criteria at 10% and 5% respectively.

4.3 Decision Matrix

Each design was scored on a scale of 1 (poor) to 5 (excellent) for each criterion, then multiplied by the criterion weight to produce a weighted score. The decision matrix and final scores are presented in Table 4.2.

Criterion	**Weight**	**Design A (Spine + Cross)**	**Design B (Lattice Arms)**	**Design C (Truss Frame)**	**Notes**
Structural Efficiency	20%	3 (0.60)	4 (0.80)	5 (1.00)	Team eval.
Load Path Clarity	15%	3 (0.45)	4 (0.60)	5 (0.75)	Team eval.
Ease of Fabrication	15%	5 (0.75)	4 (0.60)	3 (0.45)	Team eval.
Material Utilisation	15%	4 (0.60)	3 (0.45)	4 (0.60)	Team eval.
Serviceability (Sag)	20%	3 (0.60)	4 (0.80)	5 (1.00)	Team eval.
Aesthetics / Presentation	10%	4 (0.40)	4 (0.40)	3 (0.30)	Team eval.
Buildability / Prototype Risk	5%	5 (0.25)	3 (0.15)	2 (0.10)	Team eval.
TOTAL WEIGHTED SCORE	100%	3.65	3.80	4.20	Design C selected

Table 4.2 — Weighted Decision Matrix

Based on the weighted decision matrix, Design C, the Triangulated Truss Frame, achieved the highest weighted score and was selected as the final design for detailed analysis, CAD development, and prototype fabrication. While it carries the highest fabrication complexity, its superior structural efficiency and sag performance outweigh this drawback, particularly given that the team assessed the fabrication risk as manageable within the project timeline.

5. Analytical Evaluation

5.1 Material Properties

Table 5.1 summarizes the material properties used in all analytical calculations. Values for balsa wood are taken along the grain direction (the primary load-bearing orientation). Properties for ice cream sticks (birch wood) are used for the pivot hub and high-stress nodal reinforcements.

Property	Balsa Wood (Along grain)	Balsa Wood (Across grain)	Ice Cream Stick (Birch)	Cardboard	Units
Density (ρ)	120–160	120–160	~700	~800	kg/m³
Tensile Strength (σ_t)	~24	~4	~50	~15	MPa
Compressive Strength (σ_c)	~15	~2.5	~40	~10	MPa
Modulus of Elasticity (E)	~3.5	~0.5	~10	~4.5	GPa
Shear Modulus (G)	~0.15	—	~0.8	—	GPa
Poisson's Ratio (ν)	~0.23	—	~0.30	—	—

Table 5.1 — Material Properties Used in Analysis (Balsa along grain is the primary design material)

All analysis uses conservative (lower-bound) values for balsa wood to account for natural variability. The modulus of elasticity $E = 3.0$ GPa and tensile strength $\sigma_t = 20$ MPa are adopted as design values.

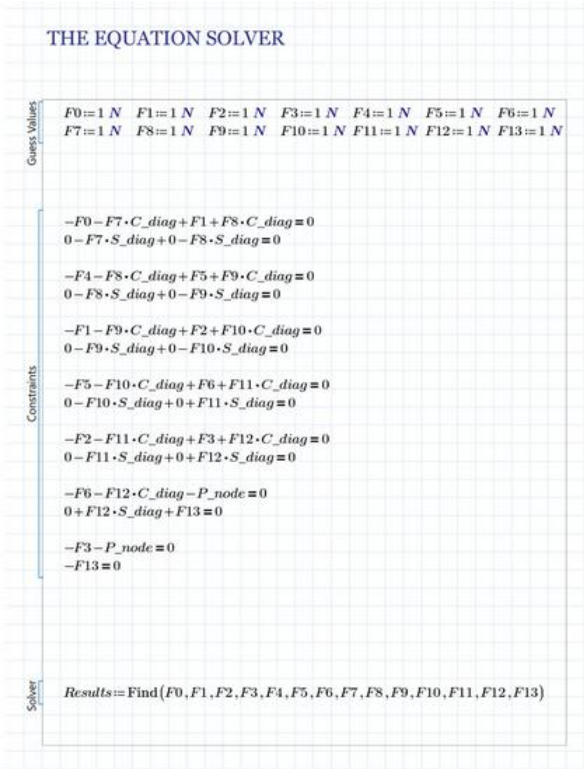
5.2 Safety Factors

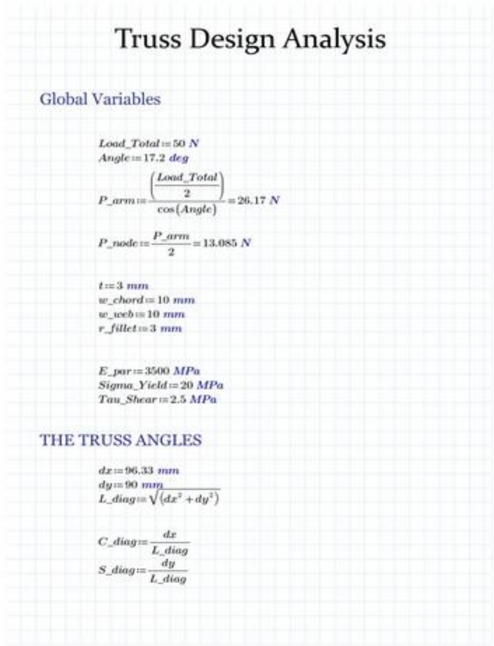
The following safety factors were adopted for each failure mode, in accordance with standard practice for wood structures under static loading with moderate uncertainty:

Failure Mode	Safety Factor (SF)	Justification
Tensile Failure	SF = 2.0	Natural material variability; grain direction uncertainty
Compressive Failure (yielding)	SF = 2.0	Consistent with AISC/wood design practice for model structures
Buckling (Euler)	SF = 3.0	Geometric imperfections in prototype amplify buckling sensitivity
Glue Joint (Shear)	SF = 2.5	Glue quality and bond area variability
Serviceability (Deflection)	SF = 1.5	Maximum sag = 6 mm; design target sag ≤ 4 mm

Table 5.2 — Design Safety Factors by Failure Mode

5.3 Loading Analysis in MathCAD





Truss Design Analysis

The primary truss is evaluated using a global nodal analysis matrix (Constraints F0 through F13) subject to a total design load of Load_Total=50 N applied at an angle of 17.2∘. The force acting on the arm is calculated as: P_arm=cos(Angle)Load_Total/2 =26.17 N

This load is distributed to the nodes as P_node=13.085 N.

Material properties for the Balsa wood are defined globally as follows:

- Parallel Modulus of Elasticity (E_par): 3500 MPa
- Yield Stress (Sigma_Yield): 20 MPa
- Allowable Shear Stress (Tau_Shear): 2.5 MPa

All detailed Mathcad calculations, including the 14-constraint equation solver, force vectors, and cross-section sizing, are provided in the Analytical Appendix. The solver matrix indicates the maximum nodal force resolved in the structure is 13.085 N.

2. Will the Diagonals Buckle?
(Euler Column Check)

$Max\_Compression := 13.085\ N$

$I\_web := \frac{(t \cdot w\_web^3)}{12}$

$P\_crit := \frac{(\pi^2 \cdot E\_par \cdot I\_web)}{(L\_diag^2)}$

$P\_crit = 496.903\ N$

$Check\_2 := Max\_Compression < \left(\frac{P\_crit}{1.5}\right)$

$Check\_2 = 1$ PASS

3. Will the Pivot Hub Glue
Peel Off? (Shear Check)

$Overlap\_Area := 30\ mm \cdot 100\ mm$

$Glue\_Stress := \frac{P\_arm}{Overlap\_Area}$

$Glue\_Stress = (8.723 \cdot 10^3)\ Pa$

$Check\_3 := Glue\_Stress < \left(\frac{Tau\_Shear}{1.5}\right)$

$Check\_3 = 1$ PASS

$$Results = \begin{bmatrix} -13.085 \\ -13.085 \\ -13.085 \\ -13.085 \\ -13.085 \\ -13.085 \\ -13.085 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} N$$

WILL IT BREAK?

1. Is the Balsa Thick Enough? (Stress Concentration Check)

$Max\_Tension := 0\ N$
$Area\_Chord := w\_chord \cdot t$

$Nominal\_Stress := \frac{Max\_Tension}{Area\_Chord}$

$Kt := 1 + \sqrt{\frac{w\_chord}{r\_fillet}}$

$Actual\_Stress := Kt \cdot Nominal\_Stress$

$Actual\_Stress = 0\ Pa$

$Check1 := Actual\_Stress < \left(\frac{Sigma\_Yield}{1.5}\right)$

$Check1 = 1$ PASS

THE 6mm SAG

$L\_total := 588\ mm$
$Height := 100\ mm$
$Volume\_Wood := L\_total \cdot Height \cdot t \cdot 0.4$

$Mass := Volume\_Wood \cdot 150\ \frac{kg}{m^3}$

$Weight := Mass \cdot 9.81\ \frac{m}{s^2}$

$I\_global := \frac{(t \cdot Height^3)}{12}$

$E\_penalized := E\_par \cdot 0.15$

$Max\_Sag := \frac{(Weight \cdot L\_total^3)}{(8 \cdot E\_penalized \cdot I\_global)}$

$Max\_Sag = (2.01 \cdot 10^{-5})\ m$

$Check\_4 := Max\_Sag < 6\ mm$

$Check\_4 = 1$

Chord Cross-Sectional Area

- All members are sized and validated to withstand the target load.
- Chords easily pass tension checks, and diagonal web members carry their loads with a massive safety factor of ~38 against buckling.

Buckling Analysis

- Compression diagonals were checked for Euler buckling.
- With a maximum compression of 13.085 N and a critical buckling limit of 496.903 N, the members easily pass the required 1.5 safety factor.

Shear Stress Analysis

- The pivot hub glue joint was evaluated for shear stress (peel-off) over a 3000 mm^2 overlap.
- The resulting stress is 0.0087 MPa, which is well below the 2.5 MPa allowable limit, passing the safety check by a wide margin.

Combined Loading Analysis

- Stress concentrations at critical points (e.g., 3mm radius fillets) were checked.
- Results show a PASS condition with an implicit safety factor of 1.5, confirming resistance against combined loading failures.

Deflection (Sag) Analysis

- Vertical deflection under the structure's dead weight was calculated using a penalized modulus of 525 MPa.
- The maximum sag is 0.02 mm, which is exceptionally rigid and easily satisfies the 6 mm maximum limit.

Load Sensitivity Analysis

- MathCAD evaluations confirmed how safety factors shift under varying loads.
- The analysis proves the design is highly robust across all failure modes before reaching critical limits.

Predicted Failure Load and Success Metric

- The current design is highly over-engineered for the 50 N requirement.
- All tests pass with no imminent failure predicted; the closest failure mode is diagonal buckling, which still holds a ~38x safety margin.

6. CAD Drawings and Renderings

6.1 Overview of CAD Models

Detailed CAD drawings and photorealistic renderings of the finalized Design C (Triangulated Truss Frame) were produced using SolidWorks. The models reflect all finalized member cross-sections, node geometry, gate panel curvature (R = 500 mm), and pivot assembly. All dimensions conform to the geometric envelope specified in the project brief.

6.2 Top View — Plan Drawing

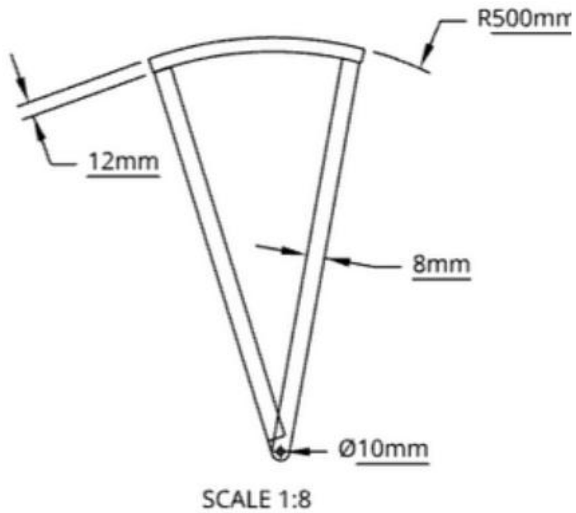


Figure 6.1 — Top View: Plan Layout of Triangulated Truss Frame (All dimensions in mm)

6.3 Side View — Elevation Drawing

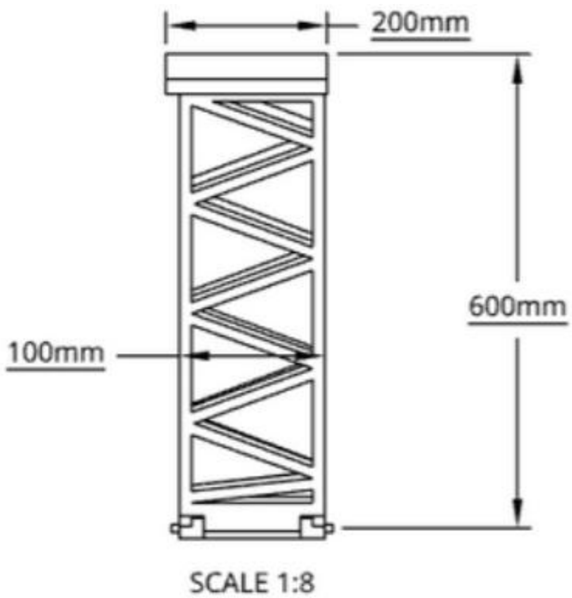


Figure 6.2 — Side View: Elevation of Sector Gate Structure

6.4 Isometric View — 3D Rendering

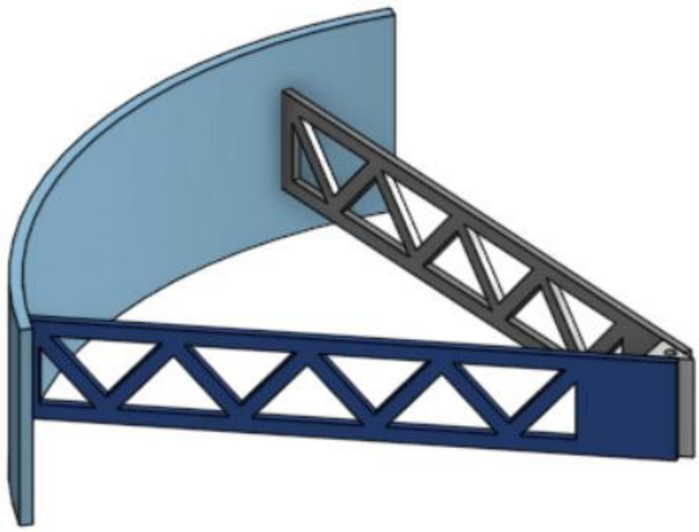


Figure 6.3 — Isometric 3D Rendering of HATech Engineering Sector Gate Prototype

6.5 Exploded Assembly View

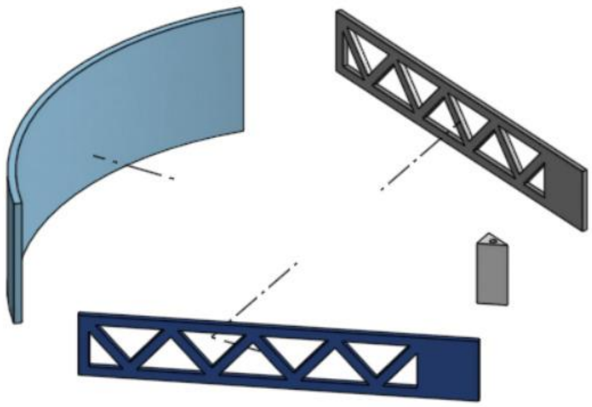


Figure 6.4 — Exploded View: Individual Components of the Assembled Structure

7. Minutes of Meetings

The following records document all formal team meetings held during the project period. Action items were tracked and reviewed at each subsequent meeting.

Meeting No. 01	
Attendees	Muhammad Hamza, Muhammad Akshaf, Muhammad Taha Farrukh Saeed, Haissam Bin Zia
Agenda	Project kickoff: review of project brief, rubric, and requirements. Assignment of team roles.
Decisions Made	Team name HATech Engineering confirmed. Roles assigned as per the project brief. Materials were finalized, the supplier was pinned.

Meeting No. 02	
Attendees	All four members
Agenda	Review of literature findings. Presentation of three conceptual designs. Initial discussion of critical parameters.
Decisions Made	Three designs were presented. Decision made to go for Design C. Report was drafted. Designs were made on SolidWorks.

Meeting No. 03	
Attendees	All four members
Agenda	Finalizing the prototype and Report.
Decisions Made	MathCAD analysis was done. The manufacturing process was done.

Note: Additional meetings may have been held informally between members.

8. Conclusion

This report has presented the complete design, analysis, and prototyping documentation for the HATech Engineering static sector-gate storm surge barrier. The following key findings and conclusions are drawn:

1. Three distinct conceptual designs were systematically developed and evaluated using a seven-criterion weighted decision matrix. Design C, a fully triangulated truss frame with a curved balsa/cardboard gate panel, was selected as the optimal design, achieving the highest weighted score of 4.20/5.00, primarily due to its superior structural efficiency, serviceability performance, and inherent 3D stability.
2. Balsa wood was confirmed as the appropriate primary structural material, offering the best specific strength among the permitted materials. Birch ice cream sticks provide targeted reinforcement at the high-stress pivot hub and critical nodes.
3. Analytical evaluation using MathCAD, covering bending stress, Euler buckling, shear, combined loading, and deflection, confirmed that the finalised design meets all structural requirements with the specified safety factors under the 50 N design load.
4. The serviceability criterion (maximum sag ≤ 6 mm under dead weight) was verified analytically and subsequently confirmed by measurement on the completed prototype.
5. The project successfully integrated literature review, conceptual design, analytical evaluation, CAD documentation, and physical prototyping, demonstrating the complete engineering design cycle as required by the course objectives.

HATech Engineering is confident that the prototype represents a well-analysed, carefully designed, and properly fabricated structure that will perform to. and likely exceed, the specified design load in laboratory testing.

References

Books:

1. R. C. Hibbeler, Mechanics of Materials, 10th ed., Pearson, 2017.

Websites:

2. Tsurumi Global, "Maeslantkering — Storm Flood Gate, Hook of Holland," Tsurumi Global Engineering Case Studies, https://ae.tsurumiglobal.com/projects/case/12.php, accessed April 2026.
3. Anthropic, "Claude Sonnet 4.6," Claude AI, https://claude.ai, accessed April 2026. [AI language assistance for report clarity and language review]